

by Victoria Memorial not knowing what that building is and its place in history, you would probably say “ah, that looks like a nice colonial palace” and stop at that. However, while travelling, we are continually looking out for information, let us say the nearest restaurant in a city unknown to you. An AR browser application (which helps you browse your locality with AR information) could visually guide you to your point of interest.

From yet another perspective, travel has for long been associated with carrying paper maps, books to read during a journey, and paper to take notes on. The mighty mobile phone has replaced most of these items. Further, the development of smartphones has reduced (if not eliminated) the need to carry multiple ‘information sources’. A single device can now encompass a GPS, camera, accelerometer, internet connectivity and a compass in our mobile phones.

AR browsers can guide you step-by-step from point A to point B without ever needing to look at a map. However most AR applications currently do not ‘understand’ what is in front of you. For instance, current AR applications do not transmit the live video feed sourced through a mobile camera to some remote server to tell you what you are looking at (such an approach would be very expensive from the computational viewpoint and also consume a lot of bandwidth). Instead, information about where you are standing and what direction your camera is facing, in the form of GPS, compass and gyro-sensor data, is sent to the backend server of the app.

However, these technologies still require several improvements in accuracy and usability. Since the application does not understand what you are seeing and depends on GPS, gyro-sensor and compass data, the unavailability of any of these services will degrade the experience of the app by a large degree. For example, if the compass in your phone accurately calibrated by 90 degrees (i.e., east becomes north and north becomes west), then depending on AR app would require you to hold the phone in front of you and walk to your left! If the gyro-sensor fails, then the app might just tell you to dig the ground to reach your destination. However, such a case would be rare. What is more common is losing the GPS signal which would make the AR browser totally useless.

Even with GPS, the information being displayed can be slightly wrong. GPS detects your position with an approximate error of anything between 10 to 20 meters. Such an error can show you wrong information when you are at crossroad where multiple roads intersect or in a historical place where